CODECURE – AI HACKATHON

TRACK A

I. Drug Toxicity Prediction (Pharmacology + AI)

Drug development frequently fails due to unexpected toxicity. Early prediction of toxic compounds can significantly reduce development costs and improve patient safety.

Participants must build a **machine learning model that predicts potential drug toxicity using chemical structure and molecular descriptor data**.

Teams should also attempt to **interpret which molecular properties contribute most strongly to toxicity risk**.

Expected Tasks

Participants should:

- Process molecular descriptor datasets
- Train ML models to predict toxicity
- Identify key structural features linked to toxicity
- Build a simple prediction interface or visualization

Suggested Datasets

i. Primary Dataset: Tox21 Dataset

Link: https://www.kaggle.com/datasets/epicskills/tox21-dataset

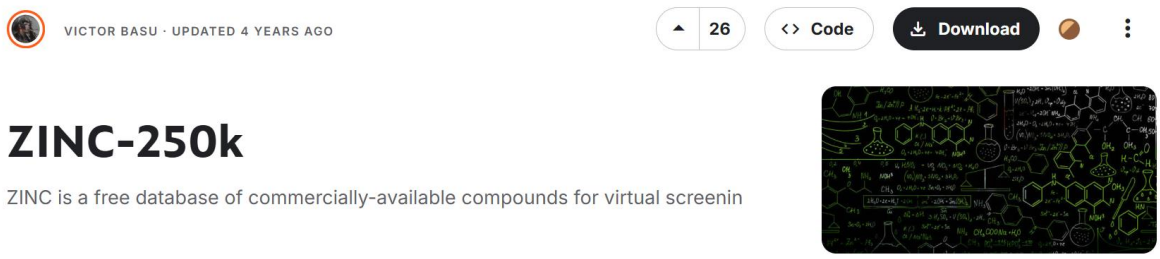
Contains:

- ~12,000 chemical compounds
- Toxicity assay results
- Molecular structures

Widely used for **AI toxicity prediction benchmarks**.

ii. Secondary Dataset: ZINC250k Molecular Dataset

Link: https://www.kaggle.com/datasets/basu369victor/zinc250k



The ZINC250k dataset is a curated subset of the ZINC compound database containing approximately 250,000 drug-like molecules commonly used in computational drug discovery and machine learning research. The dataset provides molecular structures represented as SMILES strings, along with several computed molecular properties such as logP (lipophilicity), QED (quantitative estimate of drug-likeness), and SAS (synthetic accessibility score).

Participants may use this dataset to explore chemical space, compute molecular descriptors, or engineer additional features that may improve toxicity prediction models. As this is a secondary dataset, participants are responsible for any preprocessing or feature extraction required before use.

Contains:
- ~250,000 drug-like molecules
- SMILES representations of molecular structures
- Computed molecular properties (logP, QED, SAS)
- Data suitable for molecular descriptor generation and feature engineering

Usage of this dataset is **recommended but not compulsory**.

iii. Optional Dataset: ChEMBL Bioactivity Dataset

Participants may access ChEMBL data using one of the following methods:

1. **Download ChEMBL SQLite database** (recommended for offline use)
   https://chembl.gitbook.io/chembl-interface-documentation/downloads

   For offline access to the entire database, ChEMBL provides full database dumps. This requires you to install a local database instance (e.g., PostgreSQL or MySQL) or use the self-contained SQLite version.

2. **Query the ChEMBL Web Interface** to download specific compound datasets.

3. **Use the ChEMBL Web API** to programmatically retrieve compound information.

Participants are **not required to download the entire database** and may work with smaller subsets relevant to their models.

Provides:

- drug-target activity
- molecular descriptors
- pharmacological data

Teams can use it to **augment features**. Usage is NOT COMPULSORY.

Deliverables

Teams should submit:

- GitHub Repository
- Machine learning model predicting drug toxicity
- Feature importance analysis identifying key molecular descriptors
- Visualizations showing relationships between molecular properties and toxicity (Optional)
- Simple prediction tool or interface for evaluating new compounds (Optional)

TRACK B

II. Antibiotic Resistance Prediction (Microbiology + AI)

Antimicrobial resistance is one of the most pressing global health challenges. Predicting resistance patterns can help guide treatment decisions and improve antibiotic stewardship.

Participants must develop a model that predicts antibiotic resistance based on bacterial genetic or phenotypic data.

Teams should also explore which resistance genes or features are most predictive.

Expected Tasks

Participants should:

- Analyze bacterial genome or resistance data
- Build classification models predicting resistance
- Explore feature importance
- Suggest potential treatment strategies

Suggested Datasets

i. Primary Dataset: Antimicrobial Resistance Dataset
Link: https://data.mendeley.com/datasets/ccmrx8n7mk/1

Antimicrobial Resistance Dataset

Published: 3 July 2024 | Version 1 | DOI: 10.17632/ccmrx8n7mk.1
Contributors: Bukola Atobatele, Abimbola Owoseni, Segun Adebayo

Description

The dataset details the antibiotic susceptibility of bacterial isolates collected from four locations in Osun State, Nigeria. These locations include Ede, Ife, Iwo, and Osu. Within each location, samples were obtained from three key areas: the butcher table (where beef dissection occurs), the concrete slab used for slaughtering animals, and the surrounding soil. The dataset records the response (sensitivity, intermediate, or resistance) of these isolates to five different antibiotics: Imipenem, Ceftazidime, Gentamycin, Augmentin, and Ciprofloxacin.

Download All 14 KB

Files

Dataset.xlsx 16.2 KB

This dataset contains **antibiotic susceptibility testing results for bacterial isolates**, providing information on how different bacterial species respond to commonly used antibiotics. The dataset is structured in tabular format and can be directly used for machine learning or statistical analysis tasks related to antimicrobial resistance prediction.

Includes:
• bacterial isolates from environmental or clinical samples
• antibiotics tested against each isolate
• susceptibility outcomes (e.g., Resistant, Susceptible, Intermediate)
• structured tabular data suitable for classification models

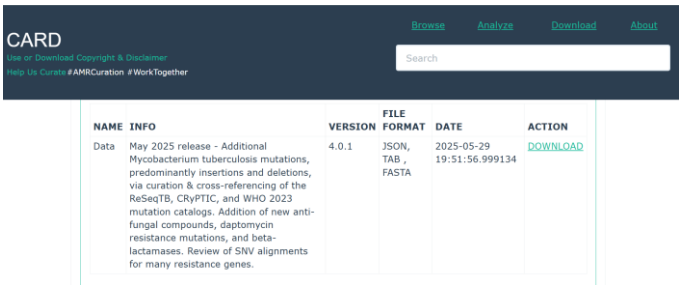
ii. Secondary Dataset: Kaggle Antibiotic Resistance Dataset
Link: https://www.kaggle.com/datasets/adilimadeddinehosni/multi-resistance-antibiotic-susceptibility

This dataset contains **bacterial isolates tested against a panel of antibiotics**, along with their corresponding susceptibility outcomes. Each record represents a bacterial strain and its response to different antimicrobial agents, enabling analysis of multi-drug resistance patterns and susceptibility trends.

Provides:
• bacterial isolate identifiers and strain information
• antibiotics tested against each isolate
• susceptibility outcomes indicating resistance or sensitivity
• resistance profiles across multiple antibiotics for multi-drug resistance analysis

This dataset can be used to **explore resistance patterns, train classification models to predict antimicrobial resistance, and analyze relationships between bacterial strains and antibiotic susceptibility**.

Optional Dataset: Comprehensive Antibiotic Resistance Database (CARD)
Link: https://card.mcmaster.ca/download



The **CARD database** is a curated resource containing detailed information about antimicrobial resistance genes and their mechanisms. Participants may optionally use this dataset to incorporate gene-level insights or biological annotations into their analysis.

Includes:
• resistance genes
• gene annotations
• antibiotic associations

Usage of this dataset is **not compulsory** and may require additional preprocessing.

Deliverables

Teams should submit:

- GitHub repository
- Resistance prediction model
- Visualization of resistance gene networks
- Decision-support tool suggesting antibiotics (Optional)

TRACK C

III. Epidemic Spread Prediction (Epidemiology + AI)

Predicting the spread of infectious diseases is essential for public health preparedness.

Participants must develop a **model that predicts disease spread or outbreak risk based on epidemiological and environmental data**.

Teams may focus on:

- outbreak prediction
- hotspot detection
- transmission modeling

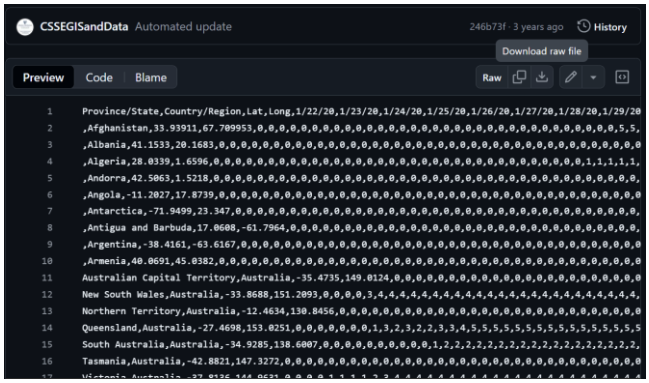
Expected Tasks

Participants should:

- analyze historical outbreak data
- build predictive models for case growth
- integrate demographic or mobility factors
- visualize predicted outbreak risk

Suggested Datasets

i. **Primary Dataset: Johns Hopkins COVID-19 Time Series Dataset**
Link: https://github.com/CSSEGISandData/COVID-19/blob/master/csse_covid_19_data/csse_covid_19_time_series/time_series_covid19_confirmed_global.csv

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```

This dataset provides **global time-series data on COVID-19 cases, deaths, and recoveries** collected by the Center for Systems Science and Engineering (CSSE) at Johns Hopkins University. It contains daily records across countries and regions and has been widely used in epidemiological modelling and machine learning studies.

Contains:
• daily confirmed COVID-19 cases across countries and regions
• daily death and recovery counts
• time-series records suitable for forecasting models
• geographic identifiers for regional outbreak analysis

This dataset can be used to **model disease spread, forecast future case counts, and identify outbreak patterns across regions**.

ii. Secondary Dataset: Our World in Data COVID-19 Dataset
Link: https://github.com/owid/covid-19-data/tree/master/public/data

covid-19-data / public / data / ↑ Top

archived	data(variants): stop sharing variants dataset (GIS...	5 years ago
cases_deaths	data(case-death): automated update	2 years ago
excess_mortality	data(xm): automated update	2 years ago
hospitalizations	data(hosp): automated update	2 years ago
internal	data(megafile): automated update	2 years ago
internal_new	data(megafile): automated update	3 years ago
jhu	data(jhu): automated update	3 years ago
latest	data(megafile): automated update	2 years ago
testing	fix(testing): manual update	4 years ago
vaccinations	data(vax): automated update	2 years ago
DATA.md	hospitalisations in catalog	2 years ago

The **Our World in Data COVID-19 dataset repository** provides a collection of global datasets related to the COVID-19 pandemic, including information on testing, vaccination, hospitalizations, mortality, and other epidemiological indicators. The repository compiles data from official national and international sources and presents them in structured tabular formats.

Participants may explore and use any relevant datasets within this repository to enrich their analysis and incorporate additional factors influencing disease spread.

Provides:
- vaccination statistics across countries and regions
- testing and healthcare indicators
- hospitalization and mortality data
- additional epidemiological variables for modelling and analysis

Usage of this dataset is **recommended but not compulsory**, and participants may select and preprocess any relevant files depending on their modelling approach.

iii. Optional Dataset: Google COVID-19 Community Mobility Reports
Link: https://www.google.com/covid19/mobility/

Community Mobility Reports

Reports created 2022-10-17.
In order to download or use the data or reports, you must agree to the Google Terms of Service.

Global CSV Region CSVs

CSV documentation How to use these reports

Search by country or region

The **Google COVID-19 Community Mobility Reports** provide anonymized insights into changes in human mobility patterns during the COVID-19 pandemic. The dataset shows how visits and length of stay at different types of places—such as workplaces, retail locations, parks, transit stations, and residential areas—changed over time compared to baseline levels.

Includes:
- regional mobility trends over time
- percentage changes in movement across different location categories
- behavioral indicators that influence disease transmission dynamics

Participants may use this dataset to **incorporate mobility patterns into their models**, analyze how population movement affects outbreak dynamics, or explore relationships between mobility changes and case growth.

Usage of this dataset is **not compulsory**, and participants may include it to enrich their epidemiological analysis if relevant to their modelling approach.

Deliverables

Teams should submit:

- GitHub Repository
- Outbreak prediction model
- Interactive epidemic dashboard
- Risk map of disease spread

This hackathon encourages participants to leverage both **traditional programming skills and modern generative AI tools** to develop innovative solutions. Participants are welcome to use generative AI systems to assist with tasks such as code generation, data preprocessing, and model development. However, the primary objective is not simply to produce code, but to demonstrate the ability to **apply computational approaches to derive biologically meaningful insights from real-world datasets**.

Teams will be evaluated on how effectively they combine **biological understanding, analytical reasoning, and computational methods** to address the problem statements. Whether through original coding, AI-assisted development, or a combination of both, the challenge ultimately tests participants' ability to **translate data into scientifically relevant interpretations and actionable insights**.